

Forder — Book IV

Proposition 18

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If AT is tangent at A to the circle through X, A, Y , and T, Y lie on opposite sides of AX , then the tangent–chord angle XAT is congruent to the inscribed angle XYA .

1 Introduction

Forder IV.18 is the direct tangent–chord theorem.

Let X, A, Y lie on a circle, let AT be tangent to the circle at A , and assume that T and Y lie on opposite sides of chord AX . Then

$$\angle XAT \cong \angle XYA.$$

In a classical treatment this theorem is often presented as the natural counterpart of IV.17. Forder’s proof is indeed extremely short: construct a ray AT' on the same side of AX as T such that

$$\angle XAT' \cong \angle XYA.$$

By IV.17 the line AT' is tangent at A . Uniqueness of the tangent then forces $AT' = AT$, and hence the required angle equality.

The formal reconstruction follows exactly this route, but it makes two points explicit that the printed proof suppresses:

1. uniqueness of the tangent *carrier line* must be proved synthetically;
2. equality of the two carrier lines is not yet enough to replace T' by T inside an angle: one must show that T, T' lie on the same ray from A .

The production implementation is

```
CGJteamLab/Proposition4_18.lean
```

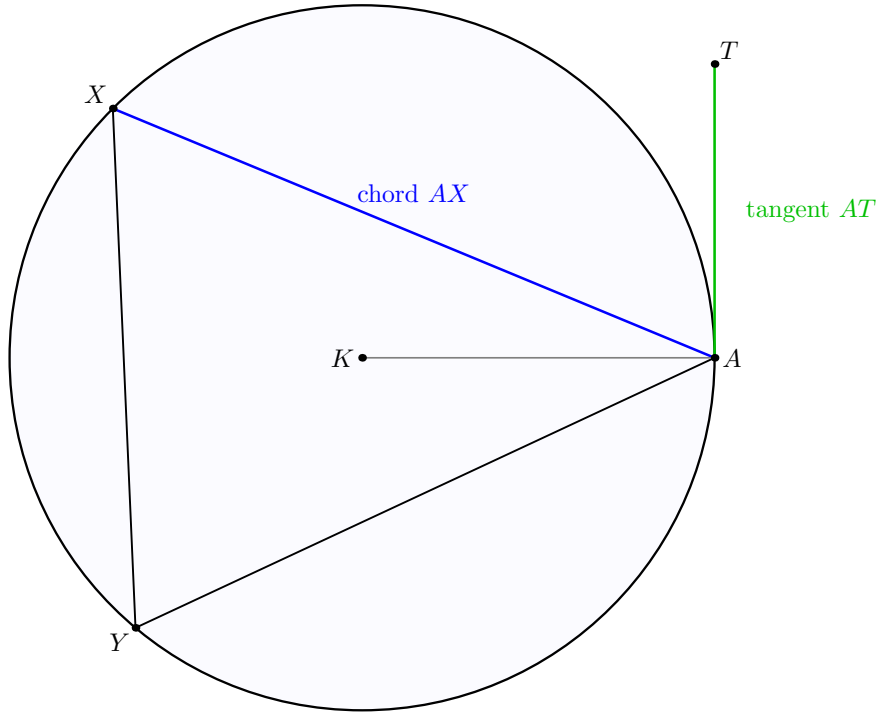
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hilbert_IV18_tangent_chord

2 Classical Configuration

The theorem starts with an actual tangent at A , not with a candidate line.

$$\angle XAT \cong \angle XYA$$



T, Y opposite sides of AX

Figure 1: Forder IV.18. The line AT is tangent at A , and T, Y lie on opposite sides of chord AX . The theorem concludes $\angle XAT \cong \angle XYA$.

The source proof introduces a second point T' , lying on the same side of the chord as T , whose ray from A is constructed to satisfy the target angle relation.

IV.17 then turns that angle relation into tangency. Thus the proof derives the direct tangent–chord theorem from its converse plus uniqueness.

3 Forder’s Statement and Proof

Forder states:

If AT is tangent to the circle XAY at A , and T, Y are on opposite sides of AX , then $\angle XAT \cong \angle XYA$.

His proof is concise. Choose T' so that

$$\angle XAT' \cong \angle XYA$$

with T, T' on the same side of AX . Then IV.17 says that AT' is tangent at A . By the uniqueness result cited by Forder, the new tangent coincides with the original one, and therefore the corresponding tangent rays determine the same angle [1].

The formal proof preserves that order exactly:

$$\text{construct } T' \implies \text{IV.17} \implies \text{tangent uniqueness} \implies \text{same ray} \implies \text{angle transport.}$$

4 Formal Interface

The proposition contract is:

```
def HilbertForderIV18TangentChord
  [H : HilbertIncidence Geo]
  [_HC : @HilbertCongruence Geo H] : Prop :=
  forall K R X A Y T : Geo.Point,
    forall chord tangent : Geo.Line,
      Ne X A ->
      H.OnLine X chord ->
      H.OnLine A chord ->
      H.OnLine A tangent ->
      H.OnLine T tangent ->
      Ne T A ->
      HilbertCircle Geo K R X ->
      HilbertCircle Geo K R A ->
      HilbertCircle Geo K R Y ->
      HilbertCircleTangentAt
        (Geo := Geo) K A tangent ->
      HilbertOppositeSide Geo T Y chord ->
      Geo.AngleCongruent
        X A T
        X Y A
```

Compared with IV.17, the hypotheses are nearly the same. The difference is precisely the direction of implication.

IV.17 assumes the angle congruence and concludes tangency.

IV.18 assumes tangency and concludes the angle congruence.

The production theorem is:

```
theorem hilbert_IV18_tangent_chord
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV18TangentChord (Geo := Geo)
```

5 Constructing the Comparison Ray

The first step is to construct a ray through A that already has the desired angle.

Because T, Y lie on opposite sides of chord AX , both are off the chord. Hence the angle

$$\angle XYA$$

is proper.

Hilbert's angle-construction theorem is applied with vertex A , first arm AX , and prescribed side equal to the side containing T .

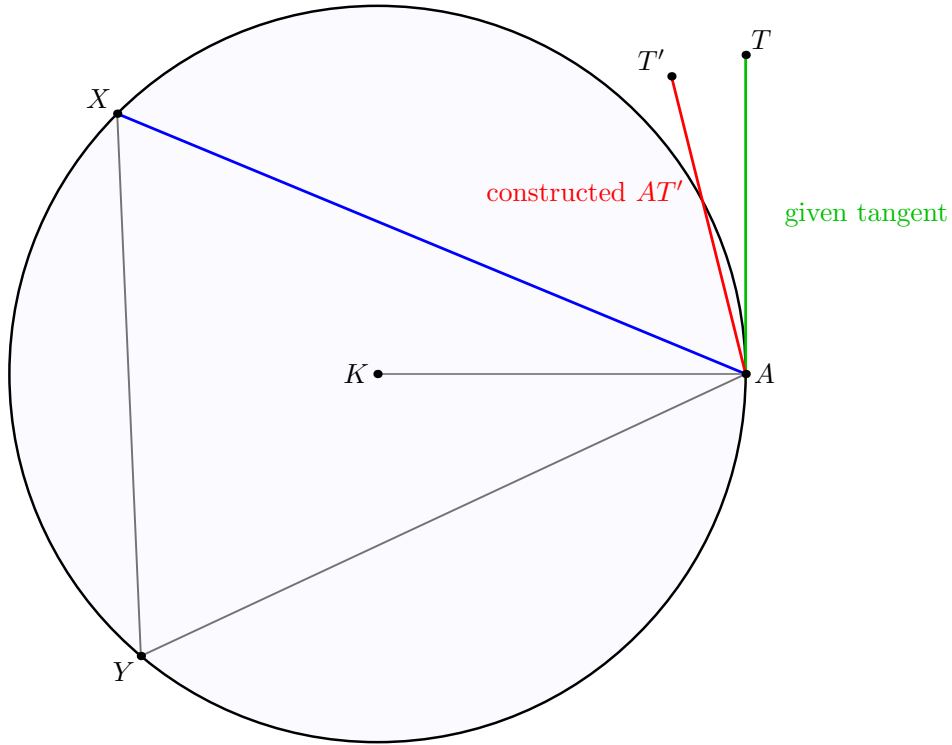
It produces a point T' such that:

$$T' \sim T$$

with respect to chord AX , and

$$\angle XAT' \cong \angle XYA.$$

T, T' same side of AX



IV.17: AT' tangent

uniqueness: carriers coincide

Figure 2: The comparison ray AT' . Hilbert angle construction places T' on the same side of chord AX as the given tangent point T and enforces $\angle XAT' \cong \angle XYA$. IV.17 will then show that AT' is itself tangent.

The formal construction is

```

rcases
  HilbertCongruence.angle_construction
    (Geo := Geo)
    X Y A
    X A T
    hXYA
    hXA
    chord
    hXchord
    hAchord
    hToff
  with
    <T', hT'TSame, hXYA_XAT', _hUnique>

```

The point T' is automatically off chord AX , because it lies on the same side of the chord as T . In particular $T' \neq A$.

6 The Side Condition for IV.17

To apply IV.17 to AT' , the constructed point T' must lie on the opposite side of chord AX from Y .

The original data give

$$T \mid Y.$$

The construction gives

$$T' \sim T.$$

Transport of opposite-side relations therefore yields

$$T' \mid Y.$$

This is a small step in the printed proof but an essential formal obligation: IV.17 requires the side relation explicitly in its interface.

7 Constructing the Carrier of AT'

The point T' determines a unique carrier line through A . Let that line be

$$\ell'.$$

The formal proof obtains it from incidence:

```

rcases
  HilbertPlaneIncidence.line_through
    A T' hAT'
  with
    <tangent', hAtangent', hT'tangent'>

```

The angle congruence from the construction is reversed into the orientation expected by IV.17:

$$\angle XAT' \cong \angle XYA.$$

All hypotheses of IV.17 are now present.

8 Using IV.17

Apply

```
hilbert_IV17_tangent_converse
```

to the new carrier ℓ' .

The conclusion is

ℓ' is tangent to the circle at A .

Thus we now have two tangent carriers at the same circle point:

ℓ and ℓ' ,

where ℓ is the original carrier containing A, T .

The remainder of the proof is a uniqueness argument.

9 Uniqueness of the Tangent Carrier

The reusable theorem is

```
hilbert_circle_tangent_carrier_unique
```

It states that if two lines through the same circle point A are tangent to the same circle, then the lines are equal.

This theorem belongs in `HilbertInterfaceIV`, not in `Proposition4_18.lean`, because it is a general theorem about the project's tangent predicate.

Its proof is itself fully synthetic.

Suppose l, m are tangent at A . Unpack the two tangent witnesses: there are points $U \in l$ and $V \in m$ such that

$$\angle UAK \quad \text{and} \quad \angle VAK$$

are right.

After swapping the angle arms, both

$$\angle KAU \quad \text{and} \quad \angle KAV$$

are right.

The reusable perpendicular-uniqueness theorem

hilbert_two_right_angles_same_first_arm_collinear

then shows that

$$U, A, V$$

are collinear.

Since A, U determine l and A, V determine m , incidence uniqueness gives

$$l = m.$$

Applied in IV.18:

$$\boxed{\ell = \ell'}.$$

10 Why Equality of Carriers Is Not Yet Enough

At this point both T and T' lie on the same tangent carrier through A .

But an angle uses a *ray*, not merely a line.

The line equality alone allows two possibilities:

$$T, T' \text{ lie on the same ray from } A,$$

or

$$A \text{ lies between } T, T'.$$

The second alternative must be excluded.

Suppose

$$T - A - T'.$$

Since A lies on chord AX , the segment TT' would meet the chord at A . This means that T, T' lie on opposite sides of chord AX .

But the angle construction deliberately placed T' on the same side of the chord as T .

Contradiction.

Hence A is not between T, T' , and the collinearity of A, T, T' therefore yields

$$\text{SameRay}(A, T, T').$$

This is the order-theoretic step that turns equality of tangent carriers into equality of tangent rays.

11 Replacing T' by T

Same-ray equality gives literal equality of the two angles:

$$\angle XAT = \angle XAT'.$$

The constructed ray already satisfies

$$\angle XAT' \cong \angle XYA.$$

Therefore

$$\boxed{\angle XAT \cong \angle XYA}.$$

This is the IV.18 conclusion.

The final Lean step is conceptually just:

```
have hAngleXAT_XAT' :
  Geo.Angle X A T =
  Geo.Angle X A T' :=
  hilbert_angle_eq_of_sameRay_second
  Geo A X T T' hRayATT'

unfold Geometry.Geo.AngleCongruent
at hXAT'_XYA |-

rw [hAngleXAT_XAT']
exact hXAT'_XYA
```

Thus the last stage of the proof contains no new circle geometry: it is ray transport.

12 Lean Formalization

The public layer is small:

```
HilbertForderIV18TangentChord
hilbert_IV18_tangent_chord
```

The proposition-specific proof is essentially the former stage-61 theorem:

```
hilbert_forder_IV18_tangent_chord_stage61
```

The general uniqueness theorem is no longer proposition-local. Its production name is

```
hilbert_circle_tangent_carrier_unique
```

in `HilbertInterfaceIV`.

The second general ingredient is

```
hilbert_two_right_angles_same_first_arm_collinear
```

which expresses the underlying uniqueness of the perpendicular through a point.

The numbered dependency is simply:

```
hilbert_IV17_tangent_converse
```

Everything else is reusable incidence, side-of-line, tangent, perpendicular, and ray infrastructure.

13 Relation to IV.17

IV.17 and IV.18 together form the tangent–chord equivalence.

IV.17 proves:

$$\angle XAT \cong \angle XYA \implies AT \text{ tangent.}$$

IV.18 proves:

$$AT \text{ tangent} \implies \angle XAT \cong \angle XYA.$$

The proof of IV.18 is deliberately asymmetric: instead of reproducing the secant analysis of IV.17 in reverse, it constructs a comparison ray and reuses IV.17.

This is a strong modularity property. The difficult secant contradiction is proved exactly once, in IV.17.

IV.18 then needs only:

$$\text{angle construction} + \text{IV.17} + \text{tangent uniqueness.}$$

14 Why Tangent Uniqueness Belongs to the Interface

During development the theorem `hilbert_circle_tangent_carrier_unique` first appeared as support for IV.18.

Its statement, however, mentions no feature specific to IV.18. It says only:

$$\text{two tangent carriers at the same circle point are equal.}$$

The production refactor therefore promotes it to `HilbertInterfaceIV`.

This ownership is important later. Any theorem that needs uniqueness of a tangent can use the interface result without importing IV.18 merely to obtain an unrelated helper.

The numbered module therefore contains the tangent–chord theorem, while the generic uniqueness principle remains proposition-independent.

15 Hilbert Components Used

15.1 Angle construction

Hilbert III.4 is the constructive heart of the proof. It produces the ray AT' realizing the angle $\angle XYA$ on a prescribed side of chord AX .

The prescribed-side clause matters: it is what later ensures T, T' are on the same ray after the tangent carriers are identified.

15.2 Plane separation

The proof uses side transport twice.

First:

$$T' \sim T, \quad T \mid Y \implies T' \mid Y.$$

Second, after tangent carrier equality, same-side information excludes

$$T - A - T'.$$

Thus plane separation enters both before and after IV.17.

15.3 Perpendicular uniqueness

The tangent predicate is defined through a right angle with the radius. Therefore uniqueness of tangent carriers ultimately reduces to uniqueness of the perpendicular to the radius through A .

15.4 Incidence uniqueness

Once the two tangent witnesses are shown collinear with A , uniqueness of the line through two distinct points identifies the two tangent carriers.

15.5 Ray equality

The final angle transport is justified only after the points T, T' are shown to lie on the same ray from A .

This distinguishes the formal proof from a looser statement that two tangent lines coincide.

16 Foundational Boundary

IV.18 introduces no new tangent axiom and no circle-specific uniqueness axiom.

The proof uses:

- Hilbert angle construction;
- plane separation;
- IV.17;
- the synthetic tangent predicate;
- uniqueness of perpendiculars;
- line uniqueness;

- ray transport.

The uniqueness of the tangent is itself derived inside the existing Hilbert Euclidean-plane framework.

No numerical angle measure and no analytic tangent construction are used.

17 Classical and Formal Dependency Structures

Forder's source proof is:

$$\begin{array}{c}
AT \text{ tangent, } T \mid Y \\
\downarrow \\
\text{construct } T' \sim T \text{ with } \angle XAT' \cong \angle XYA \\
\downarrow \\
\text{IV.17} \\
\downarrow \\
AT' \text{ tangent} \\
\downarrow \\
\text{uniqueness of tangent} \\
\downarrow \\
AT' = AT \\
\downarrow \\
\angle XAT \cong \angle XYA.
\end{array}$$

The formal proof expands the uniqueness and ray steps:

$$\begin{array}{c}
\text{Hilbert angle construction} \\
\downarrow \\
T' \sim T, \quad \angle XAT' \cong \angle XYA \\
\downarrow \\
T' \mid Y \\
\downarrow \\
\text{carrier } \ell' = AT' \\
\downarrow \\
\text{IV.17} \\
\downarrow \\
\ell' \text{ tangent} \\
\downarrow \\
\text{hilbert_circle_tangent_carrier_unique} \\
\downarrow \\
\ell' = \ell \\
\downarrow \\
A, T, T' \text{ collinear} \\
\downarrow \\
T - A - T' \text{ excluded by same-side} \\
\downarrow \\
\text{SameRay}(A, T, T') \\
\downarrow \\
\angle XAT = \angle XAT' \\
\downarrow \\
\angle XAT \cong \angle XYA.
\end{array}$$

At module level:

```
HilbertInterfaceIV
Proposition4_17
  -> Proposition4_18
```

18 Diagrammatic Audit

Two figures are sufficient for IV.18.

1. The source tangent–chord configuration.
2. The comparison construction showing the given tangent ray AT and the constructed ray AT' before uniqueness identifies their carriers.

A third figure for tangent uniqueness would largely duplicate the second figure; the logical argument is clearer in the text.

19 Conclusion

Forder IV.18 is short because almost all of its mathematical work has already been concentrated in IV.17.

The proof constructs a comparison ray AT' with the required angle, places it on the correct side of chord AX , and invokes IV.17 to prove that its carrier is tangent. A reusable tangent-carrier uniqueness theorem then identifies that carrier with the original tangent.

The formal reconstruction exposes one final distinction invisible in the source shorthand: equality of carriers does not yet give equality of rays. The same-side condition from the angle construction is used once more to exclude opposite orientation, yielding

$$\text{SameRay}(A, T, T').$$

Only then may T' be replaced by T in the angle.

Thus IV.18 is a compact but instructive example of modular proof design: construction, converse theorem, uniqueness, and ray transport combine to recover the direct tangent–chord relation without repeating the secant analysis of IV.17.

References

- [1] Henry George Forder, *Foundations of Euclidean Geometry*, Cambridge University Press, Cambridge, 1927.